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EXAMINER	
BUTTICE, AUDREY L	

ART UNIT	PAPER NUMBER
1647	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/348,706	Applicant(s) Erickson et al.	
	Examiner AUDREY L BUTTICE	Art Unit 1647	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 03 October 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-3,5-9,11-16 and 19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-3,5-9,11-16 and 19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Priority

The instant application, filed 06/15/2021, claims domestic benefit to US provisional application 63/039,146, filed 06/15/2020.

Status of Application, Amendments, and/or Claims

Applicant's response of 10/03/2025 is acknowledged. Claims 1, 6-9, 11, and 14 are amended; claims 4, 10, and 17-18 are cancelled; and claim 19 is new. Claims 1-3, 5-9, 11-16, and 19 are currently pending and are examined on the merits herein.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 07/03/2025 is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement has been considered by the examiner.

The following grounds of rejections are modified or new as necessitated by applicant's amendment to the claims.

Withdrawn Rejections

In the office action of 04/03/2025, claim 10 was rejected under 35 USC 103 and nonstatutory double patenting. The cancellation of the claim has rendered the rejections moot and the rejections are withdrawn.

Claim Interpretation

In the absence of a limiting definition of "functional resistance" in the disclosure, the term is interpreted in the instant office action as meaning resistance that shows active resistance to antibiotics in culture.

This interpretation is supported by the instant disclosure which states that PCR allows for detection of antibiotic resistance genes; however, PCR is not able to distinguish between bacteria having functional antibiotic resistance genes and those having non-functional antibiotic resistant genes, thus, resulting in false-positive detection of antibiotic-resistant bacteria. The presently embodied methods are capable of positively detecting bacteria with functional antibiotic resistance genes, without positive detection of non-functional antibiotic resistance genes. The method disclosed allows detection of functional resistance to an antibiotic even if the mechanism of resistance is not a single gene/protein or mutation (pages 32-33, [0113]).

Claim Rejections - 35 USC § 112(b)

The following is a quotation of 35 U.S.C. 112(b):

> (b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

1. **Claims 11-12 are rejected under 35 U.S.C. 112(b)** or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 11 recites the limitation "each recombinant bacteriophage of the therapeutic cocktail". There is insufficient antecedent basis for this limitation in the claim. As amended, claim 11 now depends on claim 1. Claim 1 requires that the therapeutic cocktail composition comprise at least one bacteriophage; however, the claim does not recite that the bacteriophage is a recombinant bacteriophage. As such, the recitation in claim 11 has insufficient antecedent basis.

Appropriate correction is required.

Claim 12 is rejected by virtue of its dependency on claim 11 as it does not resolve the ambiguity discussed above.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

2. **Claims 1-3, 5-9, and 11-16 are rejected under 35 U.S.C. 103** as being unpatentable over US 2019/0167736 A1 (Ferullo, D.J. and J.A. Radding) 06 JUN 2019 in view of US 2019/0100811 A1 (Belenky, A.S., et al) 04 APR 2019 and WO 2015/126966 (Anderson, D.L., et al) 27 AUG 2015.

'736 teaches that prosthetic joint infection (PJI) is often characterized by the presence of *Staphylococcus aureus* and/or other microbial strains that secrete numerous enzymes and toxins resulting in pain, and inflammation. Additionally, '736 teaches that the microbial organisms generate biofilms that can protect the organism from the host immune system and from antibiotics rendering prosthetic joint infections particularly difficult to treat (page 1, [0006]).

'736 teaches a method for determining the presence or absence of susceptible *Staphylococcus aureus* in a sample derived from a subject having prosthetic joint infection. '736 teaches that the method includes exposing the sample to a GRCS bacteriophage, or a functional derivative thereof,

where the bacteriophage is engineered to comprise a nucleic acid encoding a detectable marker, such as a luminescent or fluorescent protein, including luciferase, which is then expressed by the host bacteria. '736 teaches that the sample is subsequently assayed to detect the presence or absence of the marker, which is indicative of the presence or absence of susceptible *Staphylococcus aureus* in the sample (page 6, [0053]; page 4, [0043]). '736 teaches that the bacteriophage is engineered to comprise a nucleic acid encoding a detectable marker, such as a luminescent or fluorescent protein and provides examples of detectable proteins that can be used (page 4, [0043]).

'736 further teaches that when a sample tests positive for susceptible bacteria, the patient is then treated with the bacteriophage described in the disclosure (page 6, [0053]). '736 teaches that the therapeutic bacteriophage is engineered to encode one or more additional enzymes, which, when expressed by the target bacteria, enhance the effectiveness for clearing the infection, such as a biofilm-degrading enzyme (page 3, [0028]). '736 teaches exemplary enzymes useful for breaking down biofilms including dispersin B (page 3, [0031]) which meets the limitation of the enzyme being a glycosidase as evidenced by the instant specification page 18, [0063], which identifies dispersin B as a glycoside hydrolase enzyme. '736 further teaches that the method of treating prosthetic joint infections comprises administering the bacteriophage to the infected area and/or the surface of the prosthetic effectively inhibiting the growth of and/or killing the microorganisms involved in the infections and eliminating or reducing the bacterial biofilm produced by the microorganisms (page 5, [0048]).

'736 teaches that the bacteriophage or pharmaceutical composition of the invention may be administered in combination with an additional therapeutic agent to a subject in need thereof, including an antibiotic or antimicrobial agent, which is administered locally or systemically, and that the bacteriophage and additional therapeutic agent produces synergistic effects (page 5, [0051]). '736 teaches that biofilms may be found on any surface, including prosthetic joints, and that biofilm-degrading enzymes degrade biofilm matrix polymers by inhibiting biofilm formation, detaching

established biofilm colonies, and rendering biofilm-forming cells sensitive to killing by antimicrobial agents (page 3, [0030]).

While '736 does not explicitly state that the bacteriophages of the disclosure are recombinant, the teaching of bacteriophages engineered to comprise a nucleic acid reads on the limitations of recombinant bacteriophages based on the instant specification which defines recombinant as "genetic (i.e. nucleic acid) modifications as usually performed in a laboratory to bring together genetic material that would not otherwise be found." (page 6, [0027]).

'736 further teaches that the pharmaceutical compositions comprise one or more bacteriophages (page 5, [0047]) and that prosthetic joint infections can be a mixed infection involving *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]).

'736 differs from the instantly claimed invention in that '736 does not expressly teach the use of at least two different types of recombinant bacteriophage with different indicator proteins for testing the sample. '736 also does not teach that the method comprises the determination of antibiotic resistance, where the detection of the signal is used to determine functional resistance to the antibiotic or that the total recombinant bacteriophage concentration is greater than 1.0×10^6 PFU/mL.

'811 teaches that current approaches to identify drug resistant bacteria fail to satisfy today's need for efficient and practical means for phenotypic analysis of a large variety of bacteria, including mixtures thereof, for example, on the basis of the type of resistance that they harbor. Therefore, at the time of publication, '811 teaches that there was a need for assay systems that were useful for screening susceptibility of particular strains of bacteria to antibacterial agents (page 2, [0017]).

'811 teaches methods of identifying bacterial pathogens and defining the susceptibility of bacteria to test agents and methods for determining whether a target bacterial species is resistant to one or more antimicrobial agents (page 1, [0002]).

‘811 teaches a method for diagnosing and treating bacterial disease in a subject comprising: (a) providing a subject sample comprising a bacterial species; (b) culturing the sample to generate a plurality of primary bacterial cultures; (c) providing a plurality of transforming phages, each of which is specific to a different bacterial species, wherein each transforming phage is an engineered or recombinant phage having a gene for encoding a unique marker; (d) culturing the primary bacterial cultures of (a) in the presence of a transforming phage of (c) to provide a plurality of secondary bacteria cultures, wherein the transforming phage varies among the secondary bacteria cultures; (e) analyzing the secondary bacteria cultures to determine the presence or absence of the unique marker in the secondary cultures; (f) correlating the detection of the unique marker with a presence of a bacteria species; (g) correlating the presence of the bacterial species with the bacterial disease; and (h) optionally administering to the subject an antibiotic agent that is specific to the detected bacterial species, thereby treating the bacterial disease (page 4, [0029]).

‘811 further teaches that the methods are applicable for homogenous isolates as well as heterogenous bacteria samples and that the term “plurality” means two or more units (page 20, [0183]). ‘811 exemplifies the use of 5 types of recombinant bacteriophages (page 25, Example 3). ‘811 teaches that each engineered bacteriophage can express a unique marker, each of which has a specific activity (page 25, [0234]). ‘811 teaches that the methods are practiced using a combination of at least 2, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, or a greater number of phages (page 13, [0135]). ‘811 further teaches that the reporter molecule is a gene, referred to as a reporter gene, that encodes for expression of a detectable protein (page 17, [0145]-[0147]).

‘811 further teaches that pathogenic bacteria have acquired resistance to a majority of the antibacterial agents, thereby compromising the overall effectiveness of antibacterial therapy while also presenting new public health problems. In particular, ‘811 discusses methicillin-resistant *S. aureus*, or MRSA, which is a highly resistant pathogen and associated with nearly 94,000 new hospitalizations

annually leading to roughly 19,000 deaths/year in the US alone (page 1, [0003]). '811 teaches that the methods and compositions disclosed are useful for identifying resistance of a microbe to an antimicrobial agent or antibiotics, including cases where the phenotype of a resistant strain may include biofilm production (page 7, [0068]).

'811 teaches that methods for determining susceptibility or resistance of bacteria to an antibiotic are provided comprising antibiotic treatment, phage transformation, and detection steps and that the steps may be carried out in any order or simultaneously. In one embodiment, '811 teaches that the steps of antibiotic treatment and phage transformation are conducted sequentially (pages 20-21, [0186]).

'811 teaches a method of determining antibiotic resistance comprising culturing the bacteria in the presence or absence of an antibiotic agent to generate a plurality of primary cultures which are then cultured in the presence of transforming phages to generate secondary cultures. '811 teaches that the recombinant phages are specific to the bacteria and comprise a heterologous marker. Susceptibility is then determined by comparing the level or activity of the marker in samples that were obtained with and without the antibiotic treatment (abstract). '811 teaches that the marker is a reporter molecule that can signify its presence, e.g., its luminescent properties or its ability to conduct enzymatic reactions and teaches that a commonly used reporter gene includes luciferase (page 17, [0144]-[0146]). In this method disclosed by '811, the reporter molecule provides a signal of its presence and the levels or activity of the markers in samples of bacteria that have been treated with and without an antibiotic agent are compared and used to identify antibiotic resistance. As the resistance is determined through a comparison of signal between cultures with and without antibiotics, an ordinarily skilled artisan would reasonably recognize that the resistance is functional resistance. That is to say that the resistance determination is related to the growth impacts of the antibiotic on the bacteria in the cultures.

‘811 also teaches a list of pathogenic bacteria genera and their known host-specific bacteriophages including bacteria of the genus *staphylococcus* for which phage K is identified as a host-specific bacteriophage (page 10, [0097] – page 13, [0136]).

‘811 further teaches an exemplary workflow in which samples comprising bacteria were obtained and aliquots added to containers with and without antibiotics and incubated for a desired time at a desired temperature, for example 2 hours at 35°C (page 23, [0209]). ‘811 also teaches an example in which samples obtained from a subject were screened using recombinant bacteriophage (page 25, example 3). In the example, subjects exhibiting symptoms of bacterial infection were identified and samples were collected from cerebrospinal fluid, sputum, and blood. The samples were diluted with LB broth, thus promoting growth of all bacteria present in the respective sample, and incubated at 37°C for 4 hours (page 25, [0232]). It is noted that while ‘811 does not explicitly disclose that this is an “enrichment period”, the instant specification states that “As used herein, ‘culturing for enrichment’ refers to traditional culturing, such as incubation in media favorable to propagation of microorganisms” (page 6, [0026]). As such, the teaching of ‘811 that the samples are cultured for 2 or 4 hours prior to testing renders obvious the instant claim 15 limitation of enrichment periods of 2-24 hours.

‘966 teaches methods and systems for rapid detection of microorganisms in a sample. A modified bacteriophage is also disclosed which comprises a non-native indicator gene in the late region. The indicator product is not a fusion protein. The specificity of infectious agents allows a specific microorganism to be targeted and an indicator signal may be amplified to optimize assay sensitivity (abstract).

‘966 teaches that there is strong interest in improving speed and sensitivity for detection of bacteria, viruses, and other microorganisms in biological and clinical samples (page 1, lines 13-16) and that a need for more rapid, simple, and sensitive detection is needed (page 2, lines 10-12). ‘996 further teaches methods for detecting a microorganism of interest in a sample comprising steps of incubating

the sample with a recombinant bacteriophage that infects the microorganism of interest, wherein the bacteriophage comprises an indicator gene inserted into a late gene region of the bacteriophage such that expression of the indicator gene during bacteriophage replication following infection of the host bacteria results in a soluble indicator protein product and detecting the indicator protein product. '966 teaches that positive detection of the indicator protein indicates that the microorganism of interest is present in the sample (page 2, lines 16-25).

'966 further teaches that in some embodiments the bacteriophage concentration in the incubation step is greater than 1 x 10^7 PFU/mL (page 22, lines 29-30). Surprisingly, high concentrations of phage utilized for infecting test samples have successfully achieved detection of very low numbers of target microorganism in a very short timeframe. In some embodiments, the final bacteriophage concentration is greater than 7 x 10^6, 8 x 10^6, 7 x 10^6, 9 x 10^6, 1.0 x 10^7, 1.1 x 10^7, 1.2 x 10^7, 1.3 x 10^7, 1.4 x 10^7, 1.5 x 10^7, 1.6 x 10^7, 1.7 x 10^7, 1.8 x 10^7, 1.9 x 10^7, 2.0 x 10^7, 3.0 x 10^7, 4.0 x 10^7, 5.0 x 10^7, 6.0 x 10^7, 7.0 x 10^7, 8.0 x 10^7, 9.0 x 10^7, 1.0 x 10^8 PFU/mL (page 28, lines 15-18; and page 37, lines 14-18). '966 exemplifies the use of such concentrations of phage to successfully identify and count bacteria (pages 50-51, Example 4; page 52, Example 6; pages 52-53, Example 7; page 53, example 8, page 54, example 9; and pages 55-56, Example 11). '966 teaches that while such high concentrations were previously associated with lysis of the target cells, the "clean-up" methods disclosed for the phage stock helps to alleviate this problem (page 28, lines 19-25). '966 teaches that the use of the disclosed methods and concentrations result in detection of microorganisms in shorter timeframes and allows detection of even a single cell of the microorganism (page 37, lines 14-28).

'996 teaches that the methods solve a need by using infectious agent-based methods for amplifying a detectable signal indicating the presence of a bacteria and that in some embodiments as little as a single bacterium is detected. '996 further teaches that the principles disclosed can be applied to the detection of a variety of microorganisms (page 7, lines 3-12). '996 teaches that the indicator gene

may express a variety of biomolecules and is a gene that expresses a detectable product or an enzyme that produces a detectable product. For example, in one embodiment, the indicator gene encodes a luciferase enzyme (page 14, lines 8-12). '996 further teaches that the non-native indicator gene is under the control of a late promoter and that the use of a late gene promoter ensures that the reporter gene is not only expressed at high levels, like viral capsid proteins, but also does not shut down like endogenous bacterial genes or even early viral genes (page 14, lines 13-18).

'996 further teaches that bacterial cells detectable by the methods disclosed include pathogens of medical or veterinary significance, including *Staphylococcus aureus* (page 11, line 30 – page 12, line 3).

It would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the method of '736 to include two or more recombinant bacteriophage each with a unique marker for detection of bacteria in the sample and to determination of functional antibiotic resistance as taught by '811. It would have further been obvious to use a total recombinant bacteriophage concentration of greater than 7×10^6 PFU/mL using the methods taught by '966.

An ordinarily skilled artisan would have been motivated to use two or more bacteriophages with unique markers for detection as '736 teaches that prosthetic joint infections involve *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]) and '811 teaches that the use of a plurality of phages expressing unique markers, each of which has a specific activity, can be used for multiplex detection and heterologous bacteria samples comprising a plurality of species of bacteria. An ordinarily skilled artisan would have been motivated to determine functional antibiotic resistance as taught by '811 in the method of '736 in order to determine antibiotic susceptibility for treatment. Additionally, '736 teaches that phages can be engineered to overcome antibiotic resistance, for instance through the addition of nucleic acids encoding agents that potentiate antibiotic action and inhibiting

resistance (page 4, [0039]). An ordinarily skilled artisan would have had a reasonable expectation of success as both '736 and '811 are in the same field of using recombinant bacteriophages to identify microorganisms present in a sample demonstrating that they are analogous art.

An ordinarily skilled artisan would have been motivated to use a final bacteriophage concentration of greater than 7 x 10^6 PFU/mL and the preparation methods of '966 as '966 teaches that such concentrations of phages are able to quickly detect even a single microorganism. An ordinarily skilled artisan would have had a reasonable expectation of success as '966 provides methods of preparing phages to alleviate lysis and exemplifies the use of such concentrations and demonstrates their practicality in identifying and detecting microorganisms in a sample. Additionally, all of '736, '811, and '966 are teaching methods of detecting microorganisms of interest from samples using bacteriophages that have been genetically modified to express a detectable marker such as luciferase demonstrating that they are analogous art.

Regarding claim 3, it would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the method taught by the combination of '736, '811, and '966 to include a plurality of aliquots and reporter cocktail compositions as taught by '811. An ordinarily skilled artisan would have been motivated to use a plurality of sample aliquots and diagnostic bacteriophages as '736 teaches that prosthetic joint infections involve *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]). An ordinarily skilled artisan would have had a reasonable expectation of success as '811 demonstrates methods of determining bacterial identity and susceptibility or resistance to antibiotic or antimicrobial agents using a plurality of samples and recombinant phages.

Regarding claims 6-9, it would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the method taught by the combination of '736, '811, and '966 to have used a bacteriophage late promoter as disclosed by '966 in the construction of the bacteriophage used for detection in the method. An ordinarily skilled artisan would have been motivated to use a late promoter, in order to gain the benefits disclosed by '966 including high expression of the indicator gene. One of ordinary skill in the art would have had a reasonable expectation of success in making this combination as '736, '811, and '966 are teaching methods of detecting microorganisms of interest from samples using bacteriophages that have been genetically modified to express a detectable marker such as luciferase.

Regarding claim 14, it would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the method suggested by the combination of '736, '811, and '966 to try phage k as at least one of the bacteriophage as '811 teaches that phage k was known to infect bacteria of the genus *Staphylococcus* (US'811, page 13, [0130]). An ordinarily skilled artisan would have had a reasonable expectation of success as '736 teaches that *staphylococcus* bacteria, such as *S. aureus*, are involved in prosthetic joint infections.

Regarding claim 15, it would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the method suggested by the combination of '736, '811, and '966 to first incubate the biological sample in conditions favoring growth for an enrichment period of 2 or 4 hours as taught by '811. An ordinarily skilled artisan would have been motivated to incubate the samples in order to promote growth of all bacteria present in the respective sample, as disclosed by '811. An ordinarily skilled artisan would have had a reasonable expectation of

success as ‘811 is teaching methods of identifying and determining susceptibility of microorganisms using recombinant phages.

3. **Claim 19 is rejected under 35 U.S.C. 103** as being unpatentable over US 2019/0167736 A1 (Ferullo, D.J. and J.A. Radding) 06 JUN 2019 in view of US 2019/0100811 A1 (Belenky, A.S., et al) 04 APR 2019 and WO 2015/126966 (Anderson, D.L., et al) 27 AUG 2015 as applied to claim 1 above, and in further view of Morris, J.L., et al (2019) Evaluation of bacteriophage as an adjunct therapy for treatment of peri-prosthetic joint infection caused by *Staphylococcus aureus* PLoS ONE 14(12); e0226574.

The combination of ‘736, ‘811, and ‘966 teach the method of claim 1 as discussed in detail above.

The combination of applied references, however, does not explicitly disclose that the at least one bacteriophage of the therapeutic cocktail in the administering step is a wild-type bacteriophage.

Morris teaches that phage therapy offers a potential alternate strategy for the treatment of peri-prosthetic joint infections, particularly where limited effective antibiotics are available. Morris reports preclinical trials investigating the therapeutic efficacy of a phage cocktail, alone and in combination with vancomycin, to reduce bacterial numbers within the infected joint using a clinically-relevant model of *Staphylococcus aureus* – induced PJI (abstract).

Morris teaches that bacteriophage, or phages, are naturally occurring, obligate predators of bacteria that were discovered in the early 20th century. Phage safety in humans is well documented with phage therapy used in Eastern Europe for almost a century for the treatment of acute and chronic bacterial infections (paragraph bridging pages 1-2).

Prosthetic joint infection (PJI) following total knee arthroplasty remains the leading cause for revision surgery, with methicillin-susceptible *Staphylococcus aureus* (MSSA) the bacterium most frequently responsible. *S. aureus* biofilm formation is a key component in the virulence armamentarium

of this specific bacterium in the pathogenesis of PJI. Bio-inert orthopedic materials, such as titanium, provide habitable substrates for biofilm formation, a growth state which serves to facilitate bacterial survival in hostile environments. Not only does the structure of the biofilm limit the penetration of antibiotics and immune mediators, but recalcitrance to treatment is also driven by altered metabolic phenotype of bacterial cells within the biofilm matrix. Consequently current surgical and antibiotic management strategies for PJI are not only costly and traumatic for the patient, but also associated with considerable morbidity and mortality (page 2, paragraph 2).

In contrast to antibiotics, which decrease in concentration below the surface of bacterial biofilms, phages are capable of penetrating biofilms and self-replicating. To overcome potential limitations arising from the high phage specificity, phage cocktails have been used to broaden the spectrum of activity (page 2, paragraph 3).

Morris studied the use of five lytic *S. aureus*-specific phages, including StaPh_1, StaPh_3, StaPh_4, StaPh_11, and StaPh_16, of the family *Myoviridae*, in a StaPhage cocktail to reduce *S. aureus* numbers within biofilms (page 2, paragraph 4; page 3, paragraph 3). Morris teaches that *in vitro* spot tests were performed to confirm bactericidal activity of each phage. Individual phage suspensions contained $1x10^{9-12}$ plaque-forming units (PFU)/mL and were stored at 4°C, with titers and sterility confirmed prior to each use. For preparation of StaPhage cocktail, individual phage preparations were adjusted to $5x10^{8}$ PFU/mL in sterile 0.9% saline, then combined at equal volumes to achieve a cocktail containing $2.5x10^{9}$ PFU/mL (page 3, paragraph 3).

Morris teaches that treatment with phage alone of vancomycin alone led to 5-fold and 6.2-fold reductions, respectively, in bacterial load with peri-implant tissues compared to control animals. Compared to controls, a 22.5-fold reduction in *S. aureus* burden was observed within joint tissues of animals that were administered phage in combination with vancomycin, corresponding with decreased

swelling in the implanted knee (abstract). Morris concludes that phage therapy is a safe and effective adjunct to antibiotics for the treatment of *S. aureus* biofilm associated PJI (page 15, paragraph 3).

As Morris teaches the use of bacteriophages that are naturally occurring and of the family *Myoviridae,* an ordinarily skilled artisan would reasonably recognize that the phages in the cocktail disclosed by Morris are wild-type phages. Which is to say that they are naturally occurring and unmodified.

It would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the method taught by the combination of '736, '811, and '966 to further include the wild-type phage cocktail disclosed by Morris in the treatment of a patient that tests positive for the presence of a susceptible bacteria in the sample. An ordinarily skilled artisan would have been motivated to include the phage cocktail disclosed by Morris as Morris teaches that the use of the phage cocktail provides a 5-fold reduction in bacteria load and, when used in combination with an antibiotic, can lead to a 22.5-fold reduction in *S. aureus* burden. Additionally, Morris teaches that phage cocktails can be used to broaden the spectrum of activity. An ordinarily skilled artisan would have had a reasonable expectation of success as Morris demonstrates effective and safe use of the wild-type phage cocktail in the treatment of *S. aureus* biofilms associated with PJI, which is the same type of bacteria and biofilm infection that is taught by '736.

Response to Arguments

Applicant's arguments filed in the response of 10/03/2025 have been fully considered in so far as they apply to the modified rejections of the instant office action, but were not persuasive.

Applicant argues that '736 (referenced by applicant as "Ferullo") is not concerned with determining functional antibiotic resistance and does not suggest at least two different types of

indicator bacteriophage where the indicators provide different signal types. Applicant argues that, rather, '736 is focused on integrating the GRCS genome into a BAC. Applicant disagrees with the combination of '736 with '811 (referenced by applicant as "Belenky") and states that there is no motivation articulated for the combination in the rejection. Applicant argues that '811 does not relate to or suggest the use of bacteriophages for diagnosis and treatment of biofilm infection and does not disclose a reporter cocktail comprising at least two different types of bacteriophage. Applicant argues that '966 does not remedy this omission.

It is first noted that the amended claims do not require that the indicators provide different signal types as argued by applicant. Rather, the amended claims require that at least two different types of bacteriophage be used and that the bacteriophage express different indicator proteins, which is not necessarily the same as providing different signal types.

That said, the rejections of the instant office action are based on a combination of the applied references and what the references would have suggested to one of ordinary skill in the art prior to the effective filing date of the claimed invention. See MPEP 2145 (IV). As such, '736 is not required to teach each and every limitation of the claimed invention as '736 is not the only applied reference.

In the instant office action, '811 is applied to demonstrate that determination of functional antibiotic resistance, and the use of at least two types of phages with unique markers would have been an obvious inclusion in the methods of '736. '966 is applied in the rejection to demonstrate that the use of bacteriophage at final concentrations of more than $7x10^6$ PFU/mL would have been obvious.

While '811 and '966 do not focus on biofilm infection specifically, all of '736, '811, and '966 are teaching methods of detecting microorganisms of interest from samples using bacteriophages that have been genetically modified to express a detectable marker such as luciferase demonstrating that they are analogous art. The applied references are also in the same field of endeavor as the claimed invention which is focused on the identification of bacteria in clinical samples using recombinant bacteriophages

and are also pertinent to the problem faced by the inventor, which is the diagnosis and treatment of infections, demonstrating that they are analogous art.

MPEP 2141.01 (a).I. states “In order for a reference to be proper for use in an obviousness rejection under 35 U.S.C. 103, the reference must be analogous art to the claimed invention. In re Bigio, 381 F.3d 1320, 1325, 72 USPQ2d 1209, 1212 (Fed. Cir. 2004). A reference is analogous art to the claimed invention if: (1) the reference is from the same field of endeavor as the claimed invention (even if it addresses a different problem); or (2) the reference is reasonably pertinent to the problem faced by the inventor (even if it is not in the same field of endeavor as the claimed invention). Note that "same field of endeavor" and "reasonably pertinent" are two separate tests for establishing analogous art; it is not necessary for a reference to fulfill both tests in order to qualify as analogous art.”

Furthermore, explicit teaching, suggestion, or motivation in the art to combine the applied references is not required in order to establish a prima facie case of obviousness. MPEP 2143 provides seven exemplary rationales for supporting a conclusion of obviousness, specifically KSR (A)-(G), only one of which requires some teaching, suggestion, or motivation in the art.

That said, while ‘736 focuses on S. aureus, ‘736 teaches in multiple places that additional bacteria are likely present in the biofilm. For example, ‘736 teaches that prosthetic joint infection is characterized by the presence of S. aureus and/or other microbial strains that secrete numerous enzymes resulting in pain, inflammation, and other symptoms. These microbial organisms generate biofilms which can protect the organisms from the host immune system and from antibiotics rendering them difficult to treat (page 2, [0020]). ‘736 further teaches that S. aureus causes 20-40% of prosthetic hip and knee arthroplasty infections (page 1, [0004]) indicating that other bacteria are also involved in infections. ‘736 teaches that the prosthetic joint infection involves S. aureus and one or more additional microbial species and/or strains and that, in some embodiments, the additional strain is from coagulase-negative staphylococci, streptococci, enterococci, anaerobes, and gram-negative bacilli or other strains

of Staphylococcus such as S. epidermidis (page 5, [0050]). Therefore, while '736 teaches S. aureus, which may be known and characterized, as discussed in the rejection, an ordinarily skilled artisan would have been motivated to additionally test susceptibility and other bacteria strains present in order to identify and treat the biofilm related infection.

Nonstatutory Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

US 11,952,611 B2

Claims 1-3, 5-9, 11-16, and 19 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-21 of U.S. Patent No. 11,952,611 in view of US 2019/0167736 A1 (Ferullo, D.J. and J.A. Radding) 06 JUN 2019 in view of US 2019/0100811 A1 (Belenky, A.S., et al) 04 APR 2019, WO 2015/126966 (Anderson, D.L., et al) 27 AUG 2015, and Morris, J.L., et al (2019) Evaluation of bacteriophage as an adjunct therapy for treatment of peri-prosthetic joint infection caused by *Staphylococcus aureus* PLoS ONE 14(12); e0226574

The claims of US'611 recite a method of detecting antibiotic-resistant bacteria in a sample comprising (a) contacting the sample with an antibiotic; (b) contacting the sample with a recombinant

bacteriophage specific to the bacteria, wherein the recombinant bacteriophage comprises a genetic construct inserted into a late gene region of the bacteriophage genome, the genetic construct comprising: (i) an indicator gene, wherein the indicator gene is not continuous with a gene encoding a structural bacteriophage protein and does not yield a fusion protein; and (ii) a promoter controlling the transcription of the indicator gene, wherein the promoter is a bacteriophage late promoter, and wherein expression of the indictor gene results in an indicator protein product, and wherein the indicator protein product is luciferase; and (c) detecting a signal produced by the indicator protein product, wherein detection of the signal is used to determine the presence of antibiotic-resistant bacteria in the sample (claims 1, 17, 19, and 21). The claims further recite that the sample is contacted with the antibiotic for a period of time before contacting the infectious agent (claim 2), that the recombinant bacteriophage is derived from phage K (claim 10), and that the bacterium is *Staphylococcus aureus* (claim 11). The claims further recite that the sample is a clinical sample (claim 13). As US'611 teaches methods in which bacteria are contacted with an antibiotic and then with a phage for detection, US'611 is identifying functional antibiotic resistance.

The claims of US'611 differ from the instantly claimed invention in that US'611 does not claim that the sample is from biological sample taken from a subject to diagnose a biofilm related infection, that the detection uses two different types of recombinant bacteriophage with different indicator proteins, that the recombinant bacteriophage concentration is greater than 1.0×10^6 PFU/mL, or that the subject is treated as claimed if a biofilm-related infection is diagnosed. Based on the teachings of the prior art, however, these addition to the claims of US'611 would have been obvious.

The teachings of '736, '811, '966, and Morris are as discussed above.

It would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the claims of US'611 with the teachings of '736, '811, and '966 to arrive at the instantly claimed invention by using the methods of diagnosing and treating a

biofilm-related infection in a subject taught by '736 using two or more phage types with unique indicator proteins and a plurality of samples as disclosed by '811, and a final bacteriophage concentration is greater than 1.0×10^6 PFU/mL as disclosed by '966.

An ordinarily skilled artisan would have been motivated to use the methods claimed in US'611 for diagnosis and treatment of biofilm related infections as '736 teaches that prosthetic joint infection (PJI) is a devastating complication of prosthetic joint surgeries and that the formation of biofilms is often an impediment to effective antibiotic therapy (page 1, [0004]). An ordinarily skilled artisan would have had a reasonable expectation of success as '736 demonstrates the use of recombinant bacteriophages in the detection and treatment of such biofilm related infections. Additionally, '736 teaches that *Staphylococcus aureus* causes 20-40% of prosthetic hip and knee arthroplasty infections and US'611 claims the detection of the *Staphylococcus aureus*.

An ordinarily skilled artisan would have been motivated to use two or more phage types with unique indicator proteins and a plurality of samples as '736 teaches that prosthetic joint infections involve *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]) and '811 teaches that the use of a plurality of phages expressing unique markers, each of which has a specific activity, can be used for multiplex detection and heterologous bacteria samples comprising a plurality of species of bacteria.

An ordinarily skilled artisan would have been motivated to use greater than 7×10^6 recombinant bacteriophages and the methods of '966 as '966 teaches that such concentrations of phages are able to quickly detect even a single microorganism. An ordinarily skilled artisan would have had a reasonable expectation of success as '966 provides methods of preparing phages to alleviate lysis and exemplifies the use of such concentrations and demonstrates their practicality in identifying and detecting microorganisms in a sample. Additionally, all of US'611, '736, '811, and '966 are teaching methods of

detecting microorganisms of interest from samples using bacteriophages that have been genetically modified to express a detectable marker such as luciferase demonstrating analogous art.

Regarding claim 19, it would have further been obvious to include the wild-type phage cocktail disclosed by Morris in the treatment of a patient that tests positive for the presence of a susceptible bacteria in the sample. An ordinarily skilled artisan would have been motivated to include the phage cocktail disclosed by Morris as Morris teaches that the use of the phage cocktail provides a 5-fold reduction in bacteria load and, when used in combination with an antibiotic, can lead to a 22.5-fold reduction in *S. aureus* burden. Additionally, Morris teaches that phage cocktails can be used to broaden the spectrum of activity. An ordinarily skilled artisan would have had a reasonable expectation of success as Morris demonstrates effective and safe use of the wild-type phage cocktail in the treatment of *S. aureus* biofilms associated with PJI, which is the same type of bacteria and biofilm infection that is taught by '736.

17/534,043

Claims 1-3, 5-9, 11-16, and 19 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-8, 10-19, and 21-36 of copending Application No. 17/534,043 in view of US 2019/0167736 A1 (Ferullo, D.J. and J.A. Radding) 06 JUN 2019 in view of US 2019/0100811 A1 (Belenky, A.S., et al) 04 APR 2019, WO 2015/126966 (Anderson, D.L., et al) 27 AUG 2015, and Morris, J.L., et al (2019) Evaluation of bacteriophage as an adjunct therapy for treatment of peri-prosthetic joint infection caused by *Staphylococcus aureus* PLoS ONE 14(12); e0226574.

App'043 claims a method for the detection of a viable microorganism of interest on a surface comprising the steps of (1) obtaining a sample from the surface; (ii) incubating the sample with an indicator cocktail composition comprising at least one recombinant bacteriophage; and (iii) detecting an

indicator protein product produced by the recombinant bacteriophage, wherein positive detection of the indicator protein product indicates that the viable microorganism of interest is present in the sample (claim 1). The claims further recite that the surface is a portion of a piece of equipment, instrument, or device, including ga medical device (claims 2-3). The claims further recite that the method comprises incubating a first aliquot of the sample with a first indicator cocktail composition and incubating a second aliquot of the sample with a second indicator cocktail composition (claim 11), where the indicator cocktail composition comprises at least two recombinant bacteriophages specific for the same microorganism of interest (claim 11), which is a bacteria (claims 12-18). App'043 further claims that the microorganism is *Staphylococcus aureus, Staphylococcus lugdunensis, Klebsiella pneumonia, Klebsiella oxytoca,* or *Pseudomonas aeruginosa* (claims 15-16). The claims further recite that the recombinant bacteriophage of the indicator cocktail composition comprises a genetic construct inserted into a bacteriophage genome, where the genetic construct comprises an indicator gene and an exogenous bacteriophage late promoter (claim 24), where the indicator gene does not encode a fusion protein and the transcription of the indicator gene is controlled by the exogenous bacteriophage late promotor (claim 25). The claims further recite that the expression of the indicator gene during bacteriophage replication following infection of a host bacterium results in the indicator protein product, which is a luciferase enzyme (claims 26-27). The claims further recite determining antibiotic resistance of the detected microorganism (claim 28) where determining the antibiotic resistance comprises contacting the sample with an antibiotic prior to contacting the sample with the indicator composition (claim 29). App'043 claims that the recombinant bacteriophage is constructed from phages including phage K or MP115 (claim 30).

The claims of App'043 differ from the instantly claimed invention in that App'043 does not claim that the sample is from biological sample taken from a subject to diagnose a biofilm related infection, that the indicator composition comprises at least two bacteriophage with different indicator proteins,

that the recombinant bacteriophage concentration is greater than 1.0 x 10^6 PFU/mL, or that the subject is treated as claimed if a biofilm-related infection is diagnosed. Based on the teachings of the prior art, however, these addition to the claims of App'043 would have been obvious.

The teachings of '736, '811, '966, and Morris are as discussed above.

It would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the claims of App'043 with the teachings of '736, '811, '966, and Morris to arrive at the instantly claimed invention by using the methods of diagnosing and treating a biofilm-related infection in a subject taught by '736, using two or more phage types with unique indicator proteins and a plurality of samples as disclosed by '811, and a bacteriophage concentration is greater than 1.0 x 10^6 PFU/mL as disclosed by '966. An ordinarily skilled artisan would have been motivated to use the methods of App'043 in for diagnosis and treatment of biofilm related infections as '736 teaches that prosthetic joint infection (PJI) is a devastating complication of prosthetic joint surgeries and that the formation of biofilms is often an impediment to effective antibiotic therapy (page 1, [0004]). An ordinarily skilled artisan would have had a reasonable expectation of success as '736 demonstrates the use of recombinant bacteriophages in the detection and treatment of such biofilm related infections. Additionally, '736 teaches that *Staphylococcus aureus* causes 20-40% of prosthetic hip and knee arthroplasty infections and App'043 claims the detection of the *Staphylococcus aureus*.

An ordinarily skilled artisan would have been motivated to use two or more phage types with unique indicator proteins and a plurality of samples as '736 teaches that prosthetic joint infections involve *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]) and '811 teaches that the use of a plurality of phages expressing unique markers, each of which has a specific activity, can be used for multiplex detection and heterologous bacteria samples comprising a plurality of species of bacteria.

An ordinarily skilled artisan would have been motivated to use greater than 7 x 10^6 recombinant bacteriophages and the methods of '966 as '966 teaches that such concentrations of phages are able to quickly detect even a single microorganism. An ordinarily skilled artisan would have had a reasonable expectation of success as '966 provides methods of preparing phages to alleviate lysis and exemplifies the use of such concentrations and demonstrates their practicality in identifying and detecting microorganisms in a sample.

Regarding claim 19, it would have further been obvious to include the wild-type phage cocktail disclosed by Morris in the treatment of a patient that tests positive for the presence of a susceptible bacteria in the sample. An ordinarily skilled artisan would have been motivated to include the phage cocktail disclosed by Morris as Morris teaches that the use of the phage cocktail provides a 5-fold reduction in bacteria load and, when used in combination with an antibiotic, can lead to a 22.5-fold reduction in *S. aureus* burden. Additionally, Morris teaches that phage cocktails can be used to broaden the spectrum of activity. An ordinarily skilled artisan would have had a reasonable expectation of success as Morris demonstrates effective and safe use of the wild-type phage cocktail in the treatment of *S. aureus* biofilms associated with PJI, which is the same type of bacteria and biofilm infection that is taught by '736.

This is a provisional nonstatutory double patenting rejection.

17/050,115

Claims 1-3, 5-9, 11-16, and 19 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 16-29, 33-35, and 37-39 of copending Application No. 17/050,115 in view of US 2019/0167736 A1 (Ferullo, D.J. and J.A. Radding) 06 JUN 2019 in view of US

2019/0100811 A1 (Belenky, A.S., et al) 04 APR 2019, WO 2015/126966 (Anderson, D.L., et al) 27 AUG 2015, and Morris, J.L., et al (2019) Evaluation of bacteriophage as an adjunct therapy for treatment of peri-prosthetic joint infection caused by *Staphylococcus aureus* PLoS ONE 14(12); e0226574.

The claims of App'115 recite a method and system for detecting *Staphylococcus* in a sample comprising: incubating the sample with a recombinant bacteriophage derived from *Staphylococcus*-specific bacteriophage comprising an indicator gene inserted into a late gene region of the bacteriophage genome; and detecting an indicator protein product produced by the recombinant bacteriophage, wherein positive detection of the indicator protein indicates presence of the *Staphylococcus* in the sample (claims 1 and 26). The claims further recite that the *Staph* is *S. aureus* (claim 17) and that the sample is a clinical sample (claim 19). The claims further recite that the sample is first incubated in conditions favoring growth for an enrichment period of 9 hours or less, 8 hours or less, 7 hours or less, 6 hours or less, 5 hours or less, 4 hours or less, 3 hours or less, or 2 hours or less (claim 23).

App'115 further claims a method for selecting a treatment for a subject comprising: (i) obtaining a sample from the subject; (ii) detecting a specific antibiotic-resistant microorganism or category of microorganism in the sample using a recombinant indicator bacteriophage, where the bacteriophage has a viral late promoter, wherein detection comprises: (a) incubating the ample with at least one antibiotic to allow for enrichment of bacteria that are resistant to the antibiotic, (b) incubating the enriched samples with recombinant indicator phage where replication of the phage following infection results in production of an indicator protein product, and (c) detecting the indicator protein; and (iii) selecting a treatment based on the antibiotic resistance of the specific antibiotic resistant microorganism detected in the sample (claim 33). The claims further recite that the indicator protein is a luciferase enzyme (claim 38).

The claims of App'115 differ from the instantly claimed invention in that App'115 does not claim that the sample is from biological sample taken from a subject to diagnose a biofilm related infection, that the indicator composition comprises at least two bacteriophage with different indicator proteins, that the recombinant bacteriophage concentration is greater than 1.0×10^6 PFU/mL, or that the subject is treated as claimed if a biofilm-related infection is diagnosed. The claims of App'115 also do not explicitly claim that functional antibiotic resistance is determined. Based on the teachings of the prior art, however, these addition to the claims of App'115 would have been obvious.

The teachings of '736, '811, '966 and Morris are as discussed above.

It would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the claims of App'115 with the teachings of '736, '811, and '966 to arrive at the instantly claimed invention by using the methods of diagnosing and treating a biofilm-related infection in a subject taught by '736, using two or more phage types with unique indicator proteins, a plurality of samples, and the determination of functional resistance, as disclosed by '811, and a bacteriophage concentration is greater than 1.0×10^6 PFU/mL as disclosed by '966.

An ordinarily skilled artisan would have been motivated to use the methods of App'115 in for diagnosis and treatment of biofilm related infections as '736 teaches that prosthetic joint infection (PJI) is a devastating complication of prosthetic joint surgeries and that the formation of biofilms is often an impediment to effective antibiotic therapy (page 1, [0004]). An ordinarily skilled artisan would have had a reasonable expectation of success as '736 demonstrates the use of recombinant bacteriophages in the detection and treatment of such biofilm related infections. Additionally, '736 teaches that *Staphylococcus aureus* causes 20-40% of prosthetic hip and knee arthroplasty infections and App'115 claims the detection of the *Staphylococcus aureus*.

An ordinarily skilled artisan would have been motivated to use two or more phage types with unique indicator proteins, a plurality of samples, and methods of determining functional antibiotic

resistance as ‘736 teaches that prosthetic joint infections involve *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]) and ‘811 teaches that the use of a plurality of phages expressing unique markers, each of which has a specific activity, can be used for multiplex detection and heterologous bacteria samples comprising a plurality of species of bacteria. Additionally, an ordinarily skilled artisan would have been motivated to determine functional antibiotic resistance as taught by ‘811 in order to determine antibiotic susceptibility for treatment, particularly as ‘736 teaches that phages can be engineered to overcome antibiotic resistance, for instance through the addition of nucleic acids encoding agents that potentiate antibiotic action and inhibiting resistance (page 4, [0039]).

An ordinarily skilled artisan would have been motivated to use greater than 7×10^6 recombinant bacteriophages and the methods of ‘966 as ‘966 teaches that such concentrations of phages are able to quickly detect even a single microorganism. An ordinarily skilled artisan would have had a reasonable expectation of success as ‘966 provides methods of preparing phages to alleviate lysis and exemplifies the use of such concentrations and demonstrates their practicality in identifying and detecting microorganisms in a sample.

Regarding claim 19, it would have further been obvious to include the wild-type phage cocktail disclosed by Morris in the treatment of a patient that tests positive for the presence of a susceptible bacteria in the sample. An ordinarily skilled artisan would have been motivated to include the phage cocktail disclosed by Morris as Morris teaches that the use of the phage cocktail provides a 5-fold reduction in bacteria load and, when used in combination with an antibiotic, can lead to a 22.5-fold reduction in *S. aureus* burden. Additionally, Morris teaches that phage cocktails can be used to broaden the spectrum of activity. An ordinarily skilled artisan would have had a reasonable expectation of success as Morris demonstrates effective and safe use of the wild-type phage cocktail in the treatment

of *S. aureus* biofilms associated with PJI, which is the same type of bacteria and biofilm infection that is taught by '736.

This is a provisional nonstatutory double patenting rejection.

17/733,719

Claims 1-3, 5-9, 11-16, and 19 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1-32 of copending Application No. 17/733,719 in view of US 2019/0167736 A1 (Ferullo, D.J. and J.A. Radding) 06 JUN 2019 in view of US 2019/0100811 A1 (Belenky, A.S., et al) 04 APR 2019, WO 2015/126966 (Anderson, D.L., et al) 27 AUG 2015 and Morris, J.L., et al (2019) Evaluation of bacteriophage as an adjunct therapy for treatment of peri-prosthetic joint infection caused by *Staphylococcus aureus* PLoS ONE 14(12); e0226574.

App'719 claims a recombinant bacteriophage, a kit, a system, and a method of detecting a bacteria of interest in a sample comprising: (i) obtaining the sample; (ii) incubating the sample with at least one indicator cocktail composition, wherein the recombinant bacteriophage comprises a genetic construct inserted into the bacteriophage genome, and wherein the genetic construct comprises an indicator gene and a capture moiety encoding a capture moiety-indicator protein fusion product; (iii) capturing the capture moiety indicator protein fusion product; and (iv) detecting the captured indicator protein product, where detection of the product indicates the bacteria of interest is present (claims 1, 10, 30, and 32). The claims further recite that the bacteria of interest is a genus, species, strain or type of bacteria (claim 11), that the indicator gene encodes an enzyme (claim 13), including luciferase (claim 14), and that the bacteria includes *S. aureus*, *K. Pneumoniae*, or *P. aeruginosa* (claim 29).

The claims of App'719 differ from the instantly claimed invention in that App'719 does not claim that the sample is from biological sample taken from a subject to diagnose a biofilm related infection,

that the indicator composition comprises at least two bacteriophage with different indicator proteins, that the recombinant bacteriophage concentration is greater than 1.0 x 10^6 PFU/mL, or that the subject is treated as claimed if a biofilm-related infection is diagnosed. The claims of App'719 also do not explicitly teach that functional antibiotic resistance is determined. Based on the teachings of the prior art, however, these addition to the claims of App'719 would have been obvious.

The teachings of '736, '811, '966, and Morris are as discussed above.

It would have been *prima facie* obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify the claims of App'719 with the teachings of '736, '811, and '966 to arrive at the instantly claimed invention by using the methods of diagnosing and treating a biofilm-related infection in a subject taught by '736, using two or more phage types with unique indicator proteins, a plurality of samples, and the determination of functional resistance as disclosed by '811, and a bacteriophage concentration is greater than 1.0 x 10^6 PFU/mL as disclosed by '966.

An ordinarily skilled artisan would have been motivated to use the methods of App'719 in for diagnosis and treatment of biofilm related infections as '736 teaches that prosthetic joint infection (PJI) is a devastating complication of prosthetic joint surgeries and that the formation of biofilms is often an impediment to effective antibiotic therapy (page 1, [0004]). An ordinarily skilled artisan would have had a reasonable expectation of success as '736 demonstrates the use of recombinant bacteriophages in the detection and treatment of such biofilm related infections. Additionally, '736 teaches that *Staphylococcus aureus* causes 20-40% of prosthetic hip and knee arthroplasty infections and App'719 claims the detection of the *Staphylococcus aureus*.

An ordinarily skilled artisan would have been motivated to use two or more phage types with unique indicator proteins and a plurality of samples as '736 teaches that prosthetic joint infections involve *S. aureus* and one or more additional microbial species and/or strains (page 5, [0050]) and '811 teaches that the use of a plurality of phages expressing unique markers, each of which has a specific

activity, can be used for multiplex detection and heterologous bacteria samples comprising a plurality of species of bacteria. Additionally, an ordinarily skilled artisan would have been motivated to determine functional antibiotic resistance as taught by '811 in order to determine antibiotic susceptibility for treatment, particularly as '736 teaches that phages can be engineered to overcome antibiotic resistance, for instance through the addition of nucleic acids encoding agents that potentiate antibiotic action and inhibiting resistance (page 4, [0039]).

An ordinarily skilled artisan would have been motivated to use greater than 7×10^6 recombinant bacteriophages and the methods of '966 as '966 teaches that such concentrations of phages are able to quickly detect even a single microorganism. An ordinarily skilled artisan would have had a reasonable expectation of success as '966 provides methods of preparing phages to alleviate lysis and exemplifies the use of such concentrations and demonstrates their practicality in identifying and detecting microorganisms in a sample. Additionally, all of App'719, '736, '811, and '966 are teaching methods of detecting microorganisms of interest from samples using bacteriophages that have been genetically modified to express a detectable marker such as luciferase demonstrating analogous art.

Regarding claim 19, it would have further been obvious to include the wild-type phage cocktail disclosed by Morris in the treatment of a patient that tests positive for the presence of a susceptible bacteria in the sample. An ordinarily skilled artisan would have been motivated to include the phage cocktail disclosed by Morris as Morris teaches that the use of the phage cocktail provides a 5-fold reduction in bacteria load and, when used in combination with an antibiotic, can lead to a 22.5-fold reduction in *S. aureus* burden. Additionally, Morris teaches that phage cocktails can be used to broaden the spectrum of activity. An ordinarily skilled artisan would have had a reasonable expectation of success as Morris demonstrates effective and safe use of the wild-type phage cocktail in the treatment

of *S. aureus* biofilms associated with PJI, which is the same type of bacteria and biofilm infection that is taught by '736.

This is a provisional nonstatutory double patenting rejection.

Response to Arguments

In the remarks filed 10/03/2025, applicant requests that the nonstatutory double patenting rejections be held in abeyance until the indication of allowable subject matter.

As no allowable subject matter is indicated in the instant office action, the rejections under nonstatutory double patenting are maintained in modified form as necessitated by applicant's amendment to the claims.

Conclusion

No claims are allowed.

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to AUDREY L BUTTICE whose telephone number is (571)270-5049. The examiner can normally be reached M-Th 8:00-4:00.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Joanne Hama can be reached on 571-272-2911. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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